Working with Data

ECBS5294 · DS2 · MSBA 2026–2027

Before Block 1: what this course is, what is new, how it is graded

What this course is about

No error is not the same as correct.

- In DS1 almost everything that went wrong **crashed**: a missing file, the wrong Python, a stale kernel.
- In DS2 almost nothing crashes. The query runs, the number comes out, and **the number is wrong**: a join tripled it, a missing value vanished, the world was counted six times.
- The skill: look for evidence that a number is wrong **before anyone complains**, then build the check into the code so it fails out loud next time.

Three Mondays, three questions

Session	Date	Question	Homework
1	Mon 5 Oct	**What is this data?** — grain, keys, types, missing values; one table in SQL	HW1 due **Sat 10 Oct, 23:59**
2	Mon 12 Oct	**What happens when tables meet?** — joins; APIs and JSON into tables	HW2 due **Sat 17 Oct, 23:59**
3	Mon 19 Oct	**Can I trust the answer?** — bronze, silver, gold; checks that can fail	HW3 due **Sat 31 Oct, 23:59**

Final exam: closed-book, in person, about 70 minutes, in a separate session after HW3. The date is on Moodle. A **mock exam** with a key comes first.

Under all of it, DS1's loop: **observe → locate → hypothesize → inspect → change → verify → explain.**

Real data, every time

Data	What it is	Where you meet it
Online Retail II	525,461 invoice lines of a UK gift retailer, 2009–2010	Block 1 lecture, Lab 2, HW1
Olist	a Brazilian marketplace: 2,653 of its 99,441 orders, selected to keep every problem the labs need	Labs 3, 4, 6, HW2
Our World in Data	CO_2 and greenhouse-gas emissions, every country, every year	Lab 1, HW3
Eurostat	EU waste and emissions, exactly as Eurostat publishes them	Block 5, Lab 5, HW3
ECB via Frankfurter · World Bank	Two real APIs, cached so no lab needs the network	Block 4, Lab 4, HW2, HW3

Nothing is invented and nothing is edited. Every problem you find in a lab is something the real file does.

What is new, and what is not

The same as DS1: the rhythm (45 lecture · 5 stretch · 45 lab), the diagnosis note, **explain it to a neighbour** at minute 33, the Moodle checkpoint, labs carry no points, Git Bash on Windows.

New in DS2:

- **In the lecture you predict, then watch.** every query is on a slide; first you write down what you expect, and one reason why, then the next slide shows what it printed. **You type in the lab.** Each lab's clone starts in the stretch before it.
- **The lab has two halves:** find why a number is wrong, then write the queries the report is missing. At least one of them from an empty cell.
- **SQL on DuckDB** is the language. Every lab's `uv sync` installs it.
- **Labs 3 and 5 open with 20 minutes of no AI.**

How you are graded

Component	Weight
Homework 1 — One table, the weekly numbers	10%
Homework 2 — Many tables and an API: the board-deck numbers	15%
Homework 3 — Handover: the emissions brief	15%
Knowledge checks (start of Sessions 2 and 3, closed-book)	10%
Final exam (closed-book, in person, separate session)	50%

The same as DS1. **The exam asks you to write SQL on paper**, alone: four queries, 40 of its 100 points. The labs are the practice for it.

AI: it writes SQL that runs. Running is not right.

- **Yes:** explain this error; explain this clause; suggest what to check; improve my writing.
- **No:** submit a query or a note you cannot defend in person; describe work you did not do.
- An assistant given your schema writes a join that returns rows. **It will not tell you the rows are tripled.** The counts are yours to run.
- The habit: paste the schema and the grain, the number you expected and the number you got — and ask for the **checks** that would tell them apart, not for the fix.

Who is here

- **Eduardo Ariño de la Rubia** — RubiaE@ceu.edu · office hours by appointment.
- **Your teaching assistants** — introducing themselves now. During labs: questions, not answers.
- Every one of us has shipped a wrong number that ran without an error. Thirty seconds each.

The rules that bite this week

- **Late homework:** one day late at –10%; nothing after Sunday 23:59, because the key posts on Monday morning, before the knowledge check.
- **Extensions:** email me *before* the deadline, with a reason. Documented illness or emergencies are always accommodated.
- **Accommodations:** the skill is diagnosis, not speech. The explanation to a neighbour can be written — arrange it with me this week.
- Every other rule is the same as DS1's, and the syllabus has them all. **Moodle is authoritative** for every date.